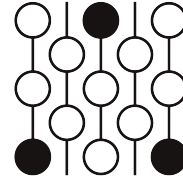


Review of Submission 5C

First Proof Second Batch

Reviewer 1

June 4–8, 2026



1 Review

Correctness. A natural approach to the problem consists of two steps: establishing the existence of an invariant measure and proving its uniqueness. The first step can be obtained relatively easily using the given approximation result. The AI successfully proves this part by using the explicit characterizations of the invariant measures for both the approximate and limiting equations.

However, in the uniqueness argument, the AI incorrectly cites a result concerning the resolvent, claiming absolute continuity with respect to an invariant measure everywhere rather than almost everywhere. This is a major mistake, and the remainder of the uniqueness proof relies on this incorrect statement.

Novelty. The submission does not appear to contain any significant new ideas. It relies heavily on results from a paper by Bounebach and Zambotti, where one can already find an explicit characterization of the invariant measure. The only part that would have required a new argument is precisely the step affected by the incorrect citation.

Overall, the submission is a compilation of cited results equipped with simple proofs.

2 Recommendation

I recommend rejection. The proof contains a major mistake by incorrectly stating an existing result. The rest of the solution does not contain any novelty.

3 Submission with inline comments

Abstract

We prove that the Markov semigroup associated with the Dirichlet stochastic heat equation with drift $\delta_0(u)$ has at most one invariant probability measure. In fact, under the hypotheses stated in the question, the invariant measure is unique and is the Gibbs perturbation of the Brownian bridge measure

$$\mu(du) = Z^{-1} \exp\left(2 \int_0^1 \mathbf{1}_{\{u(\xi) > 0\}} d\xi\right) \gamma(du),$$

where γ is the law of the standard Brownian bridge on $[0, 1]$. The proof uses three inputs: the reversible Gibbs structure of smooth approximations, the absolute-continuity theorem of Bounebaché–Zambotti for the associated Dirichlet-form resolvent, and irreducibility of the weighted Ornstein–Uhlenbeck Dirichlet form.

3.1 Notation and the candidate invariant measure

Let

$$E := C_0([0, 1]), \quad H := L^2(0, 1).$$

We regard E as a Borel subset of H . Let γ denote the law on E of the standard Brownian bridge from 0 to 0 over $[0, 1]$. Equivalently, γ is the centered Gaussian measure on H with covariance $(-\Delta_D)^{-1}$, where Δ_D is the Dirichlet Laplacian on $(0, 1)$.

Let

$$H(r) := \mathbf{1}_{\{r>0\}}, \quad \Psi(u) := \int_0^1 H(u(\xi)) \, d\xi, \quad u \in E.$$

Define

$$\mu(du) := Z^{-1} e^{2\Psi(u)} \gamma(du), \quad Z := \int_E e^{2\Psi(u)} \gamma(du). \quad (3.1)$$

Since $0 \leq \Psi \leq 1$, the measures μ and γ are equivalent and their Radon–Nikodym derivative is bounded above and below by positive constants.

For $n \geq 1$, let

$$G_{1/n}(r) = \sqrt{\frac{n}{2\pi}} e^{-nr^2/2}, \quad B_n(r) := \int_{-\infty}^r G_{1/n}(a) \, da,$$

and put

$$\Psi_n(u) := \int_0^1 B_n(u(\xi)) \, d\xi, \quad \mu_n(du) := Z_n^{-1} e^{2\Psi_n(u)} \gamma(du).$$

Then $B'_n = G_{1/n}$.

Lemma 3.1. *The measures μ_n converge to μ in total variation.*

Proof. For a Brownian bridge β ,

$$\mathbb{E}_\gamma \left[\int_0^1 \mathbf{1}_{\{\beta(\xi)=0\}} \, d\xi \right] = \int_0^1 \gamma(\beta(\xi) = 0) \, d\xi = 0,$$

because $\beta(\xi)$ is a non-degenerate centered Gaussian random variable for $\xi \in (0, 1)$, while the endpoints have Lebesgue measure zero. Hence, for γ -a.e. u , the zero set $\{\xi : u(\xi) = 0\}$ has Lebesgue measure zero. Since $B_n(r) \rightarrow H(r)$ for every $r \neq 0$ and $0 \leq B_n \leq 1$, dominated convergence gives $\Psi_n(u) \rightarrow \Psi(u)$ for γ -a.e. u . Another application of dominated convergence gives $Z_n \rightarrow Z$ and

$$Z_n^{-1} e^{2\Psi_n} \longrightarrow Z^{-1} e^{2\Psi} \quad \text{in } L^1(\gamma).$$

This is exactly total-variation convergence. □

3.2 Reversibility of the limiting semigroup

Let P_t^n be the Markov semigroup for the regularized equation

$$\partial_t u^n = \frac{1}{2} \partial_{xx} u^n + G_{1/n}(u^n) + \dot{W}$$

with Dirichlet boundary conditions. The following is standard for gradient-type stochastic heat equations with smooth drift.

Lemma 3.2. *For each n , μ_n is invariant and reversible for P_t^n :*

$$\int_E f P_t^n g \, d\mu_n = \int_E g P_t^n f \, d\mu_n$$

for all bounded Borel $f, g : E \rightarrow \mathbb{R}$ and all $t \geq 0$.

Proof. For cylinder functions φ, ψ on H , the generator of the regularized equation is

$$L_n \varphi = L_0 \varphi + \langle D\Psi_n, D\varphi \rangle_H,$$

where L_0 is the Ornstein–Uhlenbeck generator associated with the linear Dirichlet stochastic heat equation. Since $\mu_n = Z_n^{-1} e^{2\Psi_n} \gamma$, the Gaussian integration-by-parts formula gives

$$\int \psi L_n \varphi \, d\mu_n = -\frac{1}{2} \int \langle D\varphi, D\psi \rangle_H \, d\mu_n = \int \varphi L_n \psi \, d\mu_n.$$

Thus the closure of L_n is self-adjoint in $L^2(\mu_n)$, and the associated Markov semigroup is symmetric. Taking $\psi \equiv 1$ gives invariance. This is the usual Gibbs-reversibility computation for stochastic quantization; see, for example, Bounebaché–Zambotti [BZ14, Section 4] for the same calculation in the present Dirichlet setting. $\square$

We now pass to the singular limit.

Lemma 3.3. *The measure μ is invariant and reversible for P_t . Namely, for all bounded Borel $f, g : E \rightarrow \mathbb{R}$ and all $t \geq 0$,*

$$\int_E f P_t g \, d\mu = \int_E g P_t f \, d\mu. \quad (3.2)$$

Proof. First take $f, g \in C_b(E)$. By the assumed weak convergence of the regularized solutions,

$$P_t^n h(u) \longrightarrow P_t h(u), \quad u \in E,$$

for every $h \in C_b(E)$ and every fixed $t > 0$. Therefore, using Lemma 3.1,

$$\begin{aligned} \left| \int f P_t^n g \, d\mu_n - \int f P_t g \, d\mu \right| &\leq \|f\|_\infty \|g\|_\infty \|\mu_n - \mu\|_{\text{TV}} \\ &\quad + \int |f(u)| |P_t^n g(u) - P_t g(u)| \, \mu(du) \longrightarrow 0. \end{aligned}$$

Similarly,

$$\int g P_t^n f \, d\mu_n \longrightarrow \int g P_t f \, d\mu.$$

The reversibility identity for P_t^n therefore passes to the limit and yields (3.2) for bounded continuous f, g . Since E is Polish and μ is a probability measure, a monotone-class argument extends (3.2) to all bounded Borel f, g . Taking $g \equiv 1$ gives invariance of μ . $\square$

3.3 The Dirichlet form and its irreducibility

Let $\mathcal{FC}_b^1$ denote the smooth bounded cylinder functions on H . Define on $\mathcal{FC}_b^1$ the bilinear form

$$\mathcal{E}(F, G) := \frac{1}{2} \int_E \langle DF(u), DG(u) \rangle_H \mu(du). \quad (3.3)$$

Because $d\mu/d\gamma$ is bounded above and below by positive constants, (3.3) is closable in $L^2(\mu)$. We denote its closure by $(\mathcal{E}, \mathcal{F})$.

$\mathcal{F}$ is undefined

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Lemma 3.4. *The symmetric semigroup $(P_t)_{t \geq 0}$ is the Markov semigroup associated with $(\mathcal{E}, \mathcal{F})$. Moreover, $(\mathcal{E}, \mathcal{F})$ satisfies a Poincaré inequality: there exists $C < \infty$ such that for every $F \in \mathcal{F}$,*

$$\text{Var}_\mu(F) \leq C \mathcal{E}(F, F). \quad (3.4)$$

Consequently, if $F \in L^2(\mu)$ and $P_t F = F$ in $L^2(\mu)$ for all $t \geq 0$, then F is μ -a.s. constant.

Proof. For $f = -2H$, Bounebaché–Zambotti [BZ14, Propositions 2.4 and 3.1] construct the quasi-regular Dirichlet form

$$\frac{1}{2} \int \langle DF, DG \rangle d\nu_f, \quad \nu_f(du) = \frac{1}{Z_f} \exp\left(-\int_0^1 f(u(\xi)) d\xi\right) \gamma(du).$$

Since $f = -2H$, we have $\nu_f = \mu$. For smooth f , their local-time formulation is equivalent, by the occupation-time formula, to

$$\partial_t u = \frac{1}{2} \partial_{xx} u - \frac{1}{2} f'(u) + \dot{W}.$$

For $f = -2H$, one has $-\frac{1}{2}f' = \delta_0$ in the sense of distributions. Thus the Bounebaché–Zambotti process solves the same singular SPDE.

The result was cited for smooth f and then applied to a non-smooth function

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By the assumed uniqueness in law of the ABLM solution, the associated semigroup agrees with P_t . Hence P_t is the semigroup of $(\mathcal{E}, \mathcal{F})$.

It remains to prove (3.4). Let $\mathcal{E}_0$ be the Ornstein–Uhlenbeck Dirichlet form on $L^2(\gamma)$:

$$\mathcal{E}_0(F, F) = \frac{1}{2} \int \|DF\|_H^2 d\gamma.$$

The Brownian-bridge Ornstein–Uhlenbeck form has a spectral gap; equivalently, there exists $C_0 < \infty$ such that

$$\text{Var}_\gamma(F) \leq C_0 \mathcal{E}_0(F, F), \quad F \in \text{Dom}(\mathcal{E}_0).$$

This follows from the Hermite expansion of the Dirichlet Ornstein–Uhlenbeck generator; see Bounebaché–Zambotti [BZ14, Proposition 2.1].

Since $d\mu = m d\gamma$ with $0 < a \leq m \leq b < \infty$, the form domains of $\mathcal{E}$ and $\mathcal{E}_0$ coincide and

$$a \mathcal{E}_0(F, F) \leq \mathcal{E}(F, F) \leq b \mathcal{E}_0(F, F).$$

Therefore,

$$\begin{aligned} \text{Var}_\mu(F) &= \inf_{c \in \mathbb{R}} \int (F - c)^2 d\mu \\ &\leq \int \left(F - \int F d\gamma \right)^2 d\mu \\ &\leq b \text{Var}_\gamma(F) \leq bC_0 \mathcal{E}_0(F, F) \leq \frac{bC_0}{a} \mathcal{E}(F, F). \end{aligned}$$

This proves (3.4). If $P_t F = F$ for all t , then

$$\mathcal{E}(F, F) = \lim_{t \downarrow 0} \frac{1}{t} \langle F - P_t F, F \rangle_{L^2(\mu)} = 0,$$

and (3.4) implies $\text{Var}_\mu(F) = 0$. □

We also need the absolute-continuity property of the resolvent. For $\alpha > 0$, write the normalized resolvent as

$$\mathcal{R}_\alpha(x, A) := \alpha \int_0^\infty e^{-\alpha t} P_t(x, A) dt, \quad x \in E, A \in \mathcal{B}(E).$$

Lemma 3.5 (Resolvent absolute continuity). *For every $\alpha > 0$ and every $x \in E$,*

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This is a wrong statement due to an incorrect citation. The result holds for almost every $x \in E$.

$$\mathcal{R}_\alpha(x, \cdot) \ll \mu.$$

Proof. Bounebaché–Zambotti prove the absolute-continuity condition for the resolvent of the Dirichlet-form process associated with ν_f for every bounded function f of bounded variation; see [BZ14, Proposition 2.5]. Applying their result to $f = -2H$ gives absolute continuity with respect to μ . Their Proposition 3.1 identifies the associated process as a weak solution of the corresponding local-time SPDE. As explained in the proof of Lemma 3.4, for $f = -2H$ this is precisely the SPDE with drift $\delta_0(u)$. The assumed uniqueness in law of the ABLM mild solution therefore identifies this resolvent with the resolvent of P_t . Hence $\mathcal{R}_\alpha(x, \cdot) \ll \mu$. □

3.4 Uniqueness of the invariant measure

Theorem 3.6. *The semigroup $(P_t)_{t \geq 0}$ has exactly one invariant probability measure on $E = C_0([0, 1])$. It is the measure μ defined in (3.1). In particular, P_t has at most one invariant probability measure.*

Proof. By Lemma 3.3, μ is invariant. Let λ be any invariant probability measure for P_t . We first show that $\lambda \ll \mu$.

Let $A \in \mathcal{B}(E)$ satisfy $\mu(A) = 0$. By Lemma 3.5, $\mathcal{R}_\alpha(x, A) = 0$ for every $x \in E$. Since λ is invariant,

$$\begin{aligned}\lambda(A) &= \alpha \int_0^\infty e^{-\alpha t} \lambda P_t(A) \, dt \\ &= \int_E \mathcal{R}_\alpha(x, A) \lambda(\mathrm{d}x) = 0.\end{aligned}$$

Thus $\lambda \ll \mu$. Write $h := \mathrm{d}\lambda/\mathrm{d}\mu$. Then $h \geq 0$ and $\int h \, \mathrm{d}\mu = 1$.

We claim that $P_t h = h$ in $L^1(\mu)$ for every $t \geq 0$. Indeed, for every bounded Borel φ ,

$$\begin{aligned}\int_E \varphi P_t h \, \mathrm{d}\mu &= \int_E P_t \varphi h \, \mathrm{d}\mu \\ &= \int_E P_t \varphi \, \mathrm{d}\lambda \\ &= \int_E \varphi \, \mathrm{d}\lambda \\ &= \int_E \varphi h \, \mathrm{d}\mu.\end{aligned}$$

Here the first equality uses the symmetry of P_t in $L^2(\mu)$, extended by duality to $L^1(\mu)$ – $L^\infty(\mu)$, and the third equality uses invariance of λ .

Fix $N > 0$ and set $h_N := h \wedge N$. Since $r \mapsto r \wedge N$ is concave and nondecreasing, Jensen's inequality for the Markov operator P_t gives

$$P_t h_N \leq (P_t h) \wedge N = h_N.$$

Integrating with respect to the invariant measure μ ,

$$\int P_t h_N \, \mathrm{d}\mu = \int h_N \, \mathrm{d}\mu.$$

Therefore $P_t h_N = h_N$ μ -a.s. for every $t \geq 0$. Since $h_N \in L^2(\mu)$, Lemma 3.4 implies that h_N is μ -a.s. constant.

Taking $N = 2$, we obtain $h \wedge 2 = c$ for some constant c .

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The equality holds μ -a.e.

Since

$$c = \int (h \wedge 2) \, \mathrm{d}\mu \leq \int h \, \mathrm{d}\mu = 1 < 2,$$

the identity $h \wedge 2 = c$ forces $h = c$ μ -a.s. Finally, $\int h \, \mathrm{d}\mu = 1$ gives $c = 1$. Hence $\lambda = \mu$.

Thus μ is the unique invariant probability measure. $\square$

Remark 3.7. *The key point excluding additional singular invariant measures is the pointwise absolute continuity of the normalized resolvent in Lemma 3.5.*

Reviewer 1

This is again a wrong statement

Symmetry and the Poincare inequality alone imply uniqueness only among invariant probability measures that are already absolutely continuous with respect to μ .

References

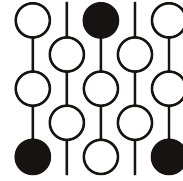
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Review of Submission 5C

First Proof Second Batch

Reviewer 2

June 4–8, 2026



1 Review

Correctness. The submission starts by using straightforward arguments to exhibit an explicit invariant measure μ . The key ingredient in the submission’s argument for the uniqueness of μ is Lemma 3.5, which shows that, pointwise in the initial condition, resolvents of the transition semigroup are absolutely continuous with respect to μ . With this claim in hand, the submission is able to reach the desired conclusion using straightforward arguments.

Lemma 3.5 is not proven in the submission, instead the submission claims Lemma 3.5 is proven in Proposition 2.5 of “A skew stochastic heat equation” by Bounebaché and Zambotti. However, upon checking the statement of this proposition in that paper, one sees that the desired absolute continuity doesn’t hold pointwise but only μ -almost surely.

The incorrect citation actually bypasses the major difficulty of the problem. In fact, in Remark 3.7, the submission itself precisely emphasizes the importance of the absolute continuity holding pointwise in Lemma 3.5.

Therefore, the paper is incomplete and is not a proof of the desired result.

Novelty. There is not any significant novelty in this submission.

2 Recommendation

The submission makes no significant progress on solving the problem nor does it have any real novelty. Therefore the submission should be **rejected**.

3 Submission with inline comments

Abstract

We prove that the Markov semigroup associated with the Dirichlet stochastic heat equation with drift $\delta_0(u)$ has at most one invariant probability measure. In fact, under the hypotheses stated in the question, the invariant measure is unique and is the Gibbs perturbation of the

Brownian bridge measure

$$\mu(du) = Z^{-1} \exp\left(2 \int_0^1 \mathbf{1}_{\{u(\xi) > 0\}} d\xi\right) \gamma(du),$$

where γ is the law of the standard Brownian bridge on $[0, 1]$. The proof uses three inputs: the reversible Gibbs structure of smooth approximations, the absolute-continuity theorem of Bounebaché–Zambotti for the associated Dirichlet-form resolvent, and irreducibility of the weighted Ornstein–Uhlenbeck Dirichlet form.

3.1 Notation and the candidate invariant measure

Let

$$E := C_0([0, 1]), \quad H := L^2(0, 1).$$

We regard E as a Borel subset of H . Let γ denote the law on E of the standard Brownian bridge from 0 to 0 over $[0, 1]$. Equivalently, γ is the centered Gaussian measure on H with covariance $(-\Delta_D)^{-1}$, where Δ_D is the Dirichlet Laplacian on $(0, 1)$.

Let

$$H(r) := \mathbf{1}_{\{r > 0\}}, \quad \Psi(u) := \int_0^1 H(u(\xi)) d\xi, \quad u \in E.$$

Define

$$\mu(du) := Z^{-1} e^{2\Psi(u)} \gamma(du), \quad Z := \int_E e^{2\Psi(u)} \gamma(du). \quad (3.1)$$

Since $0 \leq \Psi \leq 1$, the measures μ and γ are equivalent and their Radon–Nikodym derivative is bounded above and below by positive constants.

For $n \geq 1$, let

$$G_{1/n}(r) = \sqrt{\frac{n}{2\pi}} e^{-nr^2/2}, \quad B_n(r) := \int_{-\infty}^r G_{1/n}(a) da,$$

and put

$$\Psi_n(u) := \int_0^1 B_n(u(\xi)) d\xi, \quad \mu_n(du) := Z_n^{-1} e^{2\Psi_n(u)} \gamma(du).$$

Then $B'_n = G_{1/n}$.

Lemma 3.1. *The measures μ_n converge to μ in total variation.*

Proof. For a Brownian bridge β ,

$$\mathbb{E}_\gamma \left[\int_0^1 \mathbf{1}_{\{\beta(\xi)=0\}} d\xi \right] = \int_0^1 \gamma(\beta(\xi)=0) d\xi = 0,$$

because $\beta(\xi)$ is a non-degenerate centered Gaussian random variable for $\xi \in (0, 1)$, while the endpoints have Lebesgue measure zero. Hence, for γ -a.e. u , the zero set $\{\xi : u(\xi) = 0\}$ has Lebesgue measure zero. Since $B_n(r) \rightarrow H(r)$ for every $r \neq 0$ and $0 \leq B_n \leq 1$, dominated convergence gives $\Psi_n(u) \rightarrow \Psi(u)$ for γ -a.e. u . Another application of dominated convergence gives $Z_n \rightarrow Z$ and

$$Z_n^{-1} e^{2\Psi_n} \longrightarrow Z^{-1} e^{2\Psi} \quad \text{in } L^1(\gamma).$$

This is exactly total-variation convergence. □

3.2 Reversibility of the limiting semigroup

Let P_t^n be the Markov semigroup for the regularized equation

$$\partial_t u^n = \frac{1}{2} \partial_{xx} u^n + G_{1/n}(u^n) + \dot{W}$$

with Dirichlet boundary conditions. The following is standard for gradient-type stochastic heat equations with smooth drift.

Lemma 3.2. *For each n , μ_n is invariant and reversible for P_t^n :*

$$\int_E f P_t^n g \, d\mu_n = \int_E g P_t^n f \, d\mu_n$$

for all bounded Borel $f, g : E \rightarrow \mathbb{R}$ and all $t \geq 0$.

Proof. For cylinder functions φ, ψ on H , the generator of the regularized equation is

$$L_n \varphi = L_0 \varphi + \langle D\Psi_n, D\varphi \rangle_H,$$

where L_0 is the Ornstein–Uhlenbeck generator associated with the linear Dirichlet stochastic heat equation. Since $\mu_n = Z_n^{-1} e^{2\Psi_n} \gamma$, the Gaussian integration-by-parts formula gives

$$\int \psi L_n \varphi \, d\mu_n = -\frac{1}{2} \int \langle D\varphi, D\psi \rangle_H \, d\mu_n = \int \varphi L_n \psi \, d\mu_n.$$

Thus the closure of L_n is self-adjoint in $L^2(\mu_n)$, and the associated Markov semigroup is symmetric. Taking $\psi \equiv 1$ gives invariance. This is the usual Gibbs-reversibility computation for stochastic quantization; see, for example, Bounebaché–Zambotti [BZ14, Section 4] for the same calculation in the present Dirichlet setting. $\square$

We now pass to the singular limit.

Lemma 3.3. *The measure μ is invariant and reversible for P_t . Namely, for all bounded Borel $f, g : E \rightarrow \mathbb{R}$ and all $t \geq 0$,*

$$\int_E f P_t g \, d\mu = \int_E g P_t f \, d\mu. \quad (3.2)$$

Proof. First take $f, g \in C_b(E)$. By the assumed weak convergence of the regularized solutions,

$$P_t^n h(u) \longrightarrow P_t h(u), \quad u \in E,$$

for every $h \in C_b(E)$ and every fixed $t > 0$. Therefore, using Lemma 3.1,

$$\begin{aligned} \left| \int f P_t^n g \, d\mu_n - \int f P_t g \, d\mu \right| &\leq \|f\|_\infty \|g\|_\infty \|\mu_n - \mu\|_{\text{TV}} \\ &\quad + \int |f(u)| |P_t^n g(u) - P_t g(u)| \, \mu(du) \longrightarrow 0. \end{aligned}$$

Similarly,

$$\int g P_t^n f \, d\mu_n \longrightarrow \int g P_t f \, d\mu.$$

The reversibility identity for P_t^n therefore passes to the limit and yields (3.2) for bounded continuous f, g . Since E is Polish and μ is a probability measure, a monotone-class argument extends (3.2) to all bounded Borel f, g . Taking $g \equiv 1$ gives invariance of μ . $\square$

3.3 The Dirichlet form and its irreducibility

Let $\mathcal{FC}_b^1$ denote the smooth bounded cylinder functions on H . Define on $\mathcal{FC}_b^1$ the bilinear form

$$\mathcal{E}(F, G) := \frac{1}{2} \int_E \langle DF(u), DG(u) \rangle_H \mu(du). \quad (3.3)$$

Because $d\mu/d\gamma$ is bounded above and below by positive constants, (3.3) is closable in $L^2(\mu)$. We denote its closure by $(\mathcal{E}, \mathcal{F})$.

Lemma 3.4. *The symmetric semigroup $(P_t)_{t \geq 0}$ is the Markov semigroup associated with $(\mathcal{E}, \mathcal{F})$. Moreover, $(\mathcal{E}, \mathcal{F})$ satisfies a Poincaré inequality: there exists $C < \infty$ such that for every $F \in \mathcal{F}$,*

$$\text{Var}_\mu(F) \leq C \mathcal{E}(F, F). \quad (3.4)$$

Consequently, if $F \in L^2(\mu)$ and $P_t F = F$ in $L^2(\mu)$ for all $t \geq 0$, then F is μ -a.s. constant.

Proof. For $f = -2H$, Bounebaché–Zambotti [BZ14, Propositions 2.4 and 3.1] construct the quasi-regular Dirichlet form

$$\frac{1}{2} \int \langle DF, DG \rangle d\nu_f, \quad \nu_f(du) = \frac{1}{Z_f} \exp\left(-\int_0^1 f(u(\xi)) d\xi\right) \gamma(du).$$

Since $f = -2H$, we have $\nu_f = \mu$. For smooth f , their local-time formulation is equivalent, by the occupation-time formula, to

$$\partial_t u = \frac{1}{2} \partial_{xx} u - \frac{1}{2} f'(u) + \dot{W}.$$

For $f = -2H$, one has $-\frac{1}{2}f' = \delta_0$ in the sense of distributions. Thus the Bounebaché–Zambotti process solves the same singular SPDE. By the assumed uniqueness in law of the ABLM solution, the associated semigroup agrees with P_t . Hence P_t is the semigroup of $(\mathcal{E}, \mathcal{F})$.

It remains to prove (3.4). Let $\mathcal{E}_0$ be the Ornstein–Uhlenbeck Dirichlet form on $L^2(\gamma)$:

$$\mathcal{E}_0(F, F) = \frac{1}{2} \int \|DF\|_H^2 d\gamma.$$

The Brownian-bridge Ornstein–Uhlenbeck form has a spectral gap; equivalently, there exists $C_0 < \infty$ such that

$$\text{Var}_\gamma(F) \leq C_0 \mathcal{E}_0(F, F), \quad F \in \text{Dom}(\mathcal{E}_0).$$

This follows from the Hermite expansion of the Dirichlet Ornstein–Uhlenbeck generator; see Bounebaché–Zambotti [BZ14, Proposition 2.1].

Since $d\mu = m d\gamma$ with $0 < a \leq m \leq b < \infty$, the form domains of $\mathcal{E}$ and $\mathcal{E}_0$ coincide and

$$a \mathcal{E}_0(F, F) \leq \mathcal{E}(F, F) \leq b \mathcal{E}_0(F, F).$$

Therefore,

$$\begin{aligned} \text{Var}_\mu(F) &= \inf_{c \in \mathbb{R}} \int (F - c)^2 d\mu \\ &\leq \int \left(F - \int F d\gamma \right)^2 d\mu \\ &\leq b \text{Var}_\gamma(F) \leq b C_0 \mathcal{E}_0(F, F) \leq \frac{b C_0}{a} \mathcal{E}(F, F). \end{aligned}$$

This proves (3.4). If $P_t F = F$ for all t , then

$$\mathcal{E}(F, F) = \lim_{t \downarrow 0} \frac{1}{t} \langle F - P_t F, F \rangle_{L^2(\mu)} = 0,$$

and (3.4) implies $\text{Var}_\mu(F) = 0$. □

We also need the absolute-continuity property of the resolvent. For $\alpha > 0$, write the normalized resolvent as

$$\mathcal{R}_\alpha(x, A) := \alpha \int_0^\infty e^{-\alpha t} P_t(x, A) \, dt, \quad x \in E, \, A \in \mathcal{B}(E).$$

Lemma 3.5 (Resolvent absolute continuity). *For every $\alpha > 0$ and every $x \in E$,*

$$\mathcal{R}_\alpha(x, \cdot) \ll \mu.$$

Proof. Bounebaché–Zambotti prove the absolute-continuity condition for the resolvent of the Dirichlet-form process associated with ν_f for every bounded function f of bounded variation; see [BZ14, Proposition 2.5].

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[BZ14, Proposition 2.5] proves that for every $\alpha > 0$ and μ almost every $x \in E$,

$$\mathcal{R}_\alpha(x, \cdot) \ll \mu.$$

This is different from what is written in Lemma 3.5 above, and strictly weaker.

Applying their result to $f = -2H$ gives absolute continuity with respect to μ . Their Proposition 3.1 identifies the associated process as a weak solution of the corresponding local-time SPDE. As explained in the proof of Lemma 3.4, for $f = -2H$ this is precisely the SPDE with drift $\delta_0(u)$. The assumed uniqueness in law of the ABLM mild solution therefore identifies this resolvent with the resolvent of P_t . Hence $\mathcal{R}_\alpha(x, \cdot) \ll \mu$. □

3.4 Uniqueness of the invariant measure

Theorem 3.6. *The semigroup $(P_t)_{t \geq 0}$ has exactly one invariant probability measure on $E = C_0([0, 1])$. It is the measure μ defined in (3.1). In particular, P_t has at most one invariant probability measure.*

Proof. By Lemma 3.3, μ is invariant. Let λ be any invariant probability measure for P_t . We first show that $\lambda \ll \mu$.

Let $A \in \mathcal{B}(E)$ satisfy $\mu(A) = 0$. By Lemma 3.5, $\mathcal{R}_\alpha(x, A) = 0$ for every $x \in E$. Since λ is invariant,

$$\begin{aligned} \lambda(A) &= \alpha \int_0^\infty e^{-\alpha t} \lambda P_t(A) \, dt \\ &= \int_E \mathcal{R}_\alpha(x, A) \lambda(dx) = 0. \end{aligned}$$

The argument above breaks down if one uses the correct version of Lemma 3.5. Namely, knowing that $R_\alpha(x, A) = 0$ for μ -almost every x does not allow one to conclude that $\int_E \mathcal{R}_\alpha(x, A) \lambda(dx) = 0$.

Thus $\lambda \ll \mu$. Write $h := d\lambda/d\mu$. Then $h \geq 0$ and $\int h d\mu = 1$.

We claim that $P_t h = h$ in $L^1(\mu)$ for every $t \geq 0$. Indeed, for every bounded Borel φ ,

$$\begin{aligned} \int_E \varphi P_t h d\mu &= \int_E P_t \varphi h d\mu \\ &= \int_E P_t \varphi d\lambda \\ &= \int_E \varphi d\lambda \\ &= \int_E \varphi h d\mu. \end{aligned}$$

Here the first equality uses the symmetry of P_t in $L^2(\mu)$, extended by duality to $L^1(\mu)$ – $L^\infty(\mu)$, and the third equality uses invariance of λ .

Fix $N > 0$ and set $h_N := h \wedge N$. Since $r \mapsto r \wedge N$ is concave and nondecreasing, Jensen's inequality for the Markov operator P_t gives

$$P_t h_N \leq (P_t h) \wedge N = h_N.$$

Integrating with respect to the invariant measure μ ,

$$\int P_t h_N d\mu = \int h_N d\mu.$$

Therefore $P_t h_N = h_N$ μ -a.s. for every $t \geq 0$. Since $h_N \in L^2(\mu)$, Lemma 3.4 implies that h_N is μ -a.s. constant.

Taking $N = 2$, we obtain $h \wedge 2 = c$ for some constant c . Since

$$c = \int (h \wedge 2) d\mu \leq \int h d\mu = 1 < 2,$$

the identity $h \wedge 2 = c$ forces $h = c$ μ -a.s. Finally, $\int h d\mu = 1$ gives $c = 1$. Hence $\lambda = \mu$.

Thus μ is the unique invariant probability measure. $\square$

Remark 3.7. *The key point excluding additional singular invariant measures is the pointwise absolute continuity of the normalized resolvent in Lemma 3.5. Symmetry and the Poincaré inequality alone imply uniqueness only among invariant probability measures that are already absolutely continuous with respect to μ .*

References

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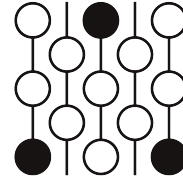
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Review of Submission 5C

First Proof Second Batch

Reviewer 3

June 4–8, 2026



1 Review

Correctness. There is a critical error in the proof of Lemma 3.5. Specifically, Proposition 2.5 of [BZ14] is quoted incorrectly. While the proof of Lemma 3.5 quotes [BZ14, Proposition 2.5] as saying that $\mathcal{R}_\alpha(x, \cdot) \ll \mu$ for every $x \in E$, in fact what the latter proposition states is that there is a measurable kernel $\rho_\alpha(x, dy)$ such that $\rho_\alpha(x, \cdot) \ll \mu$ for all $x \in E$ and $\rho_\alpha(x, \cdot) = \mathcal{R}_\alpha(x, \cdot)$ for μ -a.e. $x \in E$. The restriction of the last statement to μ -a.e. $x \in E$ means that the proposition cannot be used to prove the claim in Lemma 3.5 for *every* $x \in E$. This in turn creates a problem with the application of Lemma 3.5 in the proof of Theorem 3.6, which indeed applies Lemma 3.5 exactly for x in a set A of μ -measure 0. Thus, this error invalidates the proof of the main result.

In fact, it is known under very general conditions that two distinct ergodic invariant measures must be mutually singular. If there were another purported invariant measure ν , then one could assume by passing to an ergodic component that it is singular with respect to μ . The key point of an argument to prove the problem statement must be to obtain a contradiction to the mutual singularity of μ and ν . The submission tries to do this by showing that if an initial condition is sampled from ν , then the dynamics would take it to a random variable absolutely continuous with respect to μ , which would contradict the singularity of μ and ν by the invariance of ν . But a statement about what the dynamics does starting at μ -a.e. x is completely useless towards proving such a result, because an x sampled from ν will almost surely be in a set of μ -measure 0, which is not covered by the statement.

Indeed, in Remark 3.7 the (erroneously claimed) fact that the claim holds pointwise, not just μ -a.e., is mentioned as the “key point” in the proof.

Novelty. The key step in the proof is incorrect. The proofs of the other statements look reasonable, although I did not check them carefully. They consist of much more standard techniques.

2 Recommendation

In view of the error in the critical step of the proof mentioned above, I recommend rejection.

3 Submission with inline comments

Abstract

We prove that the Markov semigroup associated with the Dirichlet stochastic heat equation with drift $\delta_0(u)$ has at most one invariant probability measure. In fact, under the hypotheses stated in the question, the invariant measure is unique and is the Gibbs perturbation of the Brownian bridge measure

$$\mu(du) = Z^{-1} \exp\left(2 \int_0^1 \mathbf{1}_{\{u(\xi) > 0\}} d\xi\right) \gamma(du),$$

where γ is the law of the standard Brownian bridge on $[0, 1]$. The proof uses three inputs: the reversible Gibbs structure of smooth approximations, the absolute-continuity theorem of Bounebaché–Zambotti for the associated Dirichlet-form resolvent, and irreducibility of the weighted Ornstein–Uhlenbeck Dirichlet form.

3.1 Notation and the candidate invariant measure

Let

$$E := C_0([0, 1]), \quad H := L^2(0, 1).$$

We regard E as a Borel subset of H . Let γ denote the law on E of the standard Brownian bridge from 0 to 0 over $[0, 1]$. Equivalently, γ is the centered Gaussian measure on H with covariance $(-\Delta_D)^{-1}$, where Δ_D is the Dirichlet Laplacian on $(0, 1)$.

Let

$$H(r) := \mathbf{1}_{\{r > 0\}}, \quad \Psi(u) := \int_0^1 H(u(\xi)) d\xi, \quad u \in E.$$

Define

$$\mu(du) := Z^{-1} e^{2\Psi(u)} \gamma(du), \quad Z := \int_E e^{2\Psi(u)} \gamma(du). \quad (3.1)$$

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This invariant measure (along a proof of its invariance) was given in [BZ14], so this should be cited.

Since $0 \leq \Psi \leq 1$, the measures μ and γ are equivalent and their Radon–Nikodym derivative is bounded above and below by positive constants.

For $n \geq 1$, let

$$G_{1/n}(r) = \sqrt{\frac{n}{2\pi}} e^{-nr^2/2}, \quad B_n(r) := \int_{-\infty}^r G_{1/n}(a) da,$$

and put

$$\Psi_n(u) := \int_0^1 B_n(u(\xi)) d\xi, \quad \mu_n(du) := Z_n^{-1} e^{2\Psi_n(u)} \gamma(du).$$

Then $B'_n = G_{1/n}$.

Lemma 3.1. *The measures μ_n converge to μ in total variation.*

Proof. For a Brownian bridge β ,

$$\mathbb{E}_\gamma \left[\int_0^1 \mathbf{1}_{\{\beta(\xi)=0\}} d\xi \right] = \int_0^1 \gamma(\beta(\xi)=0) d\xi = 0,$$

because $\beta(\xi)$ is a non-degenerate centered Gaussian random variable for $\xi \in (0, 1)$, while the endpoints have Lebesgue measure zero. Hence, for γ -a.e. u , the zero set $\{\xi : u(\xi) = 0\}$ has Lebesgue measure zero. Since $B_n(r) \rightarrow H(r)$ for every $r \neq 0$ and $0 \leq B_n \leq 1$, dominated convergence gives $\Psi_n(u) \rightarrow \Psi(u)$ for γ -a.e. u . Another application of dominated convergence gives $Z_n \rightarrow Z$ and

$$Z_n^{-1} e^{2\Psi_n} \longrightarrow Z^{-1} e^{2\Psi} \quad \text{in } L^1(\gamma).$$

This is exactly total-variation convergence. □

3.2 Reversibility of the limiting semigroup

Let P_t^n be the Markov semigroup for the regularized equation

$$\partial_t u^n = \frac{1}{2} \partial_{xx} u^n + G_{1/n}(u^n) + \dot{W}$$

with Dirichlet boundary conditions. The following is standard for gradient-type stochastic heat equations with smooth drift.

Lemma 3.2. *For each n , μ_n is invariant and reversible for P_t^n :*

$$\int_E f P_t^n g d\mu_n = \int_E g P_t^n f d\mu_n$$

for all bounded Borel $f, g : E \rightarrow \mathbb{R}$ and all $t \geq 0$.

Proof. For cylinder functions φ, ψ on H , the generator of the regularized equation is

$$L_n \varphi = L_0 \varphi + \langle D\Psi_n, D\varphi \rangle_H,$$

where L_0 is the Ornstein–Uhlenbeck generator associated with the linear Dirichlet stochastic heat equation. Since $\mu_n = Z_n^{-1} e^{2\Psi_n} \gamma$, the Gaussian integration-by-parts formula gives

$$\int \psi L_n \varphi d\mu_n = -\frac{1}{2} \int \langle D\varphi, D\psi \rangle_H d\mu_n = \int \varphi L_n \psi d\mu_n.$$

Thus the closure of L_n is self-adjoint in $L^2(\mu_n)$, and the associated Markov semigroup is symmetric. Taking $\psi \equiv 1$ gives invariance. This is the usual Gibbs-reversibility computation for stochastic quantization; see, for example, Bounebaché–Zambotti [BZ14, Section 4] for the same calculation in the present Dirichlet setting. □

We now pass to the singular limit.

Lemma 3.3. *The measure μ is invariant and reversible for P_t . Namely, for all bounded Borel $f, g : E \rightarrow \mathbb{R}$ and all $t \geq 0$,*

$$\int_E f P_t g \, d\mu = \int_E g P_t f \, d\mu. \quad (3.2)$$

Proof. First take $f, g \in C_b(E)$. By the assumed weak convergence of the regularized solutions,

$$P_t^n h(u) \longrightarrow P_t h(u), \quad u \in E,$$

for every $h \in C_b(E)$ and every fixed $t > 0$. Therefore, using Lemma 3.1,

$$\begin{aligned} \left| \int f P_t^n g \, d\mu_n - \int f P_t g \, d\mu \right| &\leq \|f\|_\infty \|g\|_\infty \|\mu_n - \mu\|_{\text{TV}} \\ &\quad + \int |f(u)| |P_t^n g(u) - P_t g(u)| \mu(du) \longrightarrow 0. \end{aligned}$$

Similarly,

$$\int g P_t^n f \, d\mu_n \longrightarrow \int g P_t f \, d\mu.$$

The reversibility identity for P_t^n therefore passes to the limit and yields (3.2) for bounded continuous f, g . Since E is Polish and μ is a probability measure, a monotone-class argument extends (3.2) to all bounded Borel f, g . Taking $g \equiv 1$ gives invariance of μ . $\square$

3.3 The Dirichlet form and its irreducibility

Let $\mathcal{FC}_b^1$ denote the smooth bounded cylinder functions on H . Define on $\mathcal{FC}_b^1$ the bilinear form

$$\mathcal{E}(F, G) := \frac{1}{2} \int_E \langle DF(u), DG(u) \rangle_H \mu(du). \quad (3.3)$$

Because $d\mu/d\gamma$ is bounded above and below by positive constants, (3.3) is closable in $L^2(\mu)$. We denote its closure by $(\mathcal{E}, \mathcal{F})$.

Lemma 3.4. *The symmetric semigroup $(P_t)_{t \geq 0}$ is the Markov semigroup associated with $(\mathcal{E}, \mathcal{F})$. Moreover, $(\mathcal{E}, \mathcal{F})$ satisfies a Poincaré inequality: there exists $C < \infty$ such that for every $F \in \mathcal{F}$,*

$$\text{Var}_\mu(F) \leq C \mathcal{E}(F, F). \quad (3.4)$$

Consequently, if $F \in L^2(\mu)$ and $P_t F = F$ in $L^2(\mu)$ for all $t \geq 0$, then F is μ -a.s. constant.

Proof. For $f = -2H$, Bounebaché–Zambotti [BZ14, Propositions 2.4 and 3.1] construct the quasi-regular Dirichlet form

$$\frac{1}{2} \int \langle DF, DG \rangle \, d\nu_f, \quad \nu_f(du) = \frac{1}{Z_f} \exp\left(-\int_0^1 f(u(\xi)) \, d\xi\right) \gamma(du).$$

Since $f = -2H$, we have $\nu_f = \mu$. For smooth f , their local-time formulation is equivalent, by the occupation-time formula, to

$$\partial_t u = \frac{1}{2} \partial_{xx} u - \frac{1}{2} f'(u) + \dot{W}.$$

For $f = -2H$, one has $-\frac{1}{2}f' = \delta_0$ in the sense of distributions. Thus the Bounebaché–Zambotti process solves the same singular SPDE. By the assumed uniqueness in law of the ABLM solution, the associated semigroup agrees with P_t . Hence P_t is the semigroup of $(\mathcal{E}, \mathcal{F})$.

It remains to prove (3.4). Let $\mathcal{E}_0$ be the Ornstein–Uhlenbeck Dirichlet form on $L^2(\gamma)$:

$$\mathcal{E}_0(F, F) = \frac{1}{2} \int \|DF\|_H^2 d\gamma.$$

The Brownian-bridge Ornstein–Uhlenbeck form has a spectral gap; equivalently, there exists $C_0 < \infty$ such that

$$\text{Var}_\gamma(F) \leq C_0 \mathcal{E}_0(F, F), \quad F \in \text{Dom}(\mathcal{E}_0).$$

This follows from the Hermite expansion of the Dirichlet Ornstein–Uhlenbeck generator; see Bounebaché–Zambotti [BZ14, Proposition 2.1].

Since $d\mu = m d\gamma$ with $0 < a \leq m \leq b < \infty$, the form domains of $\mathcal{E}$ and $\mathcal{E}_0$ coincide and

$$a \mathcal{E}_0(F, F) \leq \mathcal{E}(F, F) \leq b \mathcal{E}_0(F, F).$$

Therefore,

$$\begin{aligned} \text{Var}_\mu(F) &= \inf_{c \in \mathbb{R}} \int (F - c)^2 d\mu \\ &\leq \int \left(F - \int F d\gamma \right)^2 d\mu \\ &\leq b \text{Var}_\gamma(F) \leq bC_0 \mathcal{E}_0(F, F) \leq \frac{bC_0}{a} \mathcal{E}(F, F). \end{aligned}$$

This proves (3.4). If $P_t F = F$ for all t , then

$$\mathcal{E}(F, F) = \lim_{t \downarrow 0} \frac{1}{t} \langle F - P_t F, F \rangle_{L^2(\mu)} = 0,$$

and (3.4) implies $\text{Var}_\mu(F) = 0$. □

We also need the absolute-continuity property of the resolvent. For $\alpha > 0$, write the normalized resolvent as

$$\mathcal{R}_\alpha(x, A) := \alpha \int_0^\infty e^{-\alpha t} P_t(x, A) dt, \quad x \in E, A \in \mathcal{B}(E).$$

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The correct version of this lemma is for μ -almost every $x \in E$.

Lemma 3.5 (Resolvent absolute continuity). *For every $\alpha > 0$ and every $x \in E$,*

$$\mathcal{R}_\alpha(x, \cdot) \ll \mu.$$

Proof. Bounebaché–Zambotti prove the absolute-continuity condition for the resolvent of the Dirichlet-form process associated with ν_f for every bounded function f of bounded variation; see [BZ14, Proposition 2.5].

[BZ14] only proves the absolute continuity for μ -a.e. x .

Applying their result to $f = -2H$ gives absolute continuity with respect to μ . Their Proposition 3.1 identifies the associated process as a weak solution of the corresponding local-time SPDE. As explained in the proof of Lemma 3.4, for $f = -2H$ this is precisely the SPDE with drift $\delta_0(u)$. The assumed uniqueness in law of the ABLM mild solution therefore identifies this resolvent with the resolvent of P_t . Hence $\mathcal{R}_\alpha(x, \cdot) \ll \mu$. $\square$

3.4 Uniqueness of the invariant measure

Theorem 3.6. *The semigroup $(P_t)_{t \geq 0}$ has exactly one invariant probability measure on $E = C_0([0, 1])$. It is the measure μ defined in (3.1). In particular, P_t has at most one invariant probability measure.*

Proof. By Lemma 3.3, μ is invariant. Let λ be any invariant probability measure for P_t . We first show that $\lambda \ll \mu$.

Let $A \in \mathcal{B}(E)$ satisfy $\mu(A) = 0$. By Lemma 3.5, $\mathcal{R}_\alpha(x, A) = 0$ for every $x \in E$.

As noted above, the proof of Lemma 3.5 only shows this for μ -a.e. $x \in E$.

Since λ is invariant,

$$\begin{aligned} \lambda(A) &= \alpha \int_0^\infty e^{-\alpha t} \lambda P_t(A) \, dt \\ &= \int_E \mathcal{R}_\alpha(x, A) \lambda(dx) = 0. \end{aligned}$$

Lemma 3.5 only proves that $\mathcal{R}_\alpha(x, A) = 0$ for μ -a.e. $x \in E$. In the case when λ is singular with respect to μ , the last integral $\int_E \mathcal{R}_\alpha(x, A) \lambda(dx)$ can be thought of as taken over a set of μ -measure 0 but λ -measure 1, as restricting to such a set does not change the value of the integral. But since this set has μ -measure 0, knowing that $\mathcal{R}_\alpha(x, A) = 0$ for μ -a.e. x constitutes no progress towards showing that $\int_E \mathcal{R}_\alpha(x, A) \lambda(dx) = 0$.

Thus $\lambda \ll \mu$. Write $h := d\lambda/d\mu$. Then $h \geq 0$ and $\int h \, d\mu = 1$.

We claim that $P_t h = h$ in $L^1(\mu)$ for every $t \geq 0$. Indeed, for every bounded Borel φ ,

$$\begin{aligned} \int_E \varphi P_t h \, d\mu &= \int_E P_t \varphi h \, d\mu \\ &= \int_E P_t \varphi \, d\lambda \\ &= \int_E \varphi \, d\lambda \\ &= \int_E \varphi h \, d\mu. \end{aligned}$$

Here the first equality uses the symmetry of P_t in $L^2(\mu)$, extended by duality to $L^1(\mu)$ – $L^\infty(\mu)$, and the third equality uses invariance of λ .

Fix $N > 0$ and set $h_N := h \wedge N$. Since $r \mapsto r \wedge N$ is concave and nondecreasing, Jensen's inequality for the Markov operator P_t gives

$$P_t h_N \leq (P_t h) \wedge N = h_N.$$

Integrating with respect to the invariant measure μ ,

$$\int P_t h_N \, d\mu = \int h_N \, d\mu.$$

Therefore $P_t h_N = h_N$ μ -a.s. for every $t \geq 0$. Since $h_N \in L^2(\mu)$, Lemma 3.4 implies that h_N is μ -a.s. constant.

Taking $N = 2$, we obtain $h \wedge 2 = c$ for some constant c . Since

$$c = \int (h \wedge 2) \, d\mu \leq \int h \, d\mu = 1 < 2,$$

the identity $h \wedge 2 = c$ forces $h = c$ μ -a.s. Finally, $\int h \, d\mu = 1$ gives $c = 1$. Hence $\lambda = \mu$.

Thus μ is the unique invariant probability measure. □

Remark 3.7. *The key point excluding additional singular invariant measures is the pointwise absolute continuity of the normalized resolvent in Lemma 3.5. Symmetry and the Poincaré inequality alone imply uniqueness only among invariant probability measures that are already absolutely continuous with respect to μ .*

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This “key point” is exactly the erroneous claim in Lemma 3.5!

References

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